

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

September 2018

EC245/207.01/207 Control Systems

Date: 24.09.2018, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## **Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## **SECTION – I**

**Q - 1(a)** Define Following parameters.

[07]

- 1) Steady state response
- 2) Unit impulse function
- 3) Rise time
- 4) Delay Time
- 5) Damping Frequency
- 6) Transient period
- 7) Peak overshoot

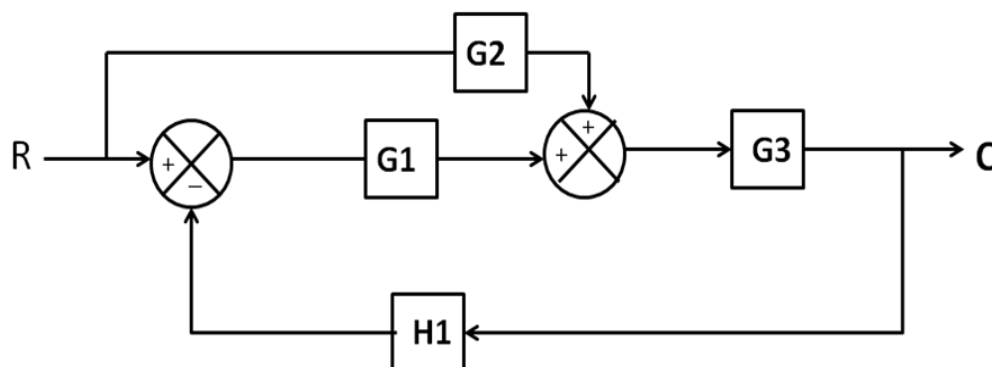
**Q - 2 (a)** Discuss the difference between open loop and closed loop control system.

[04]

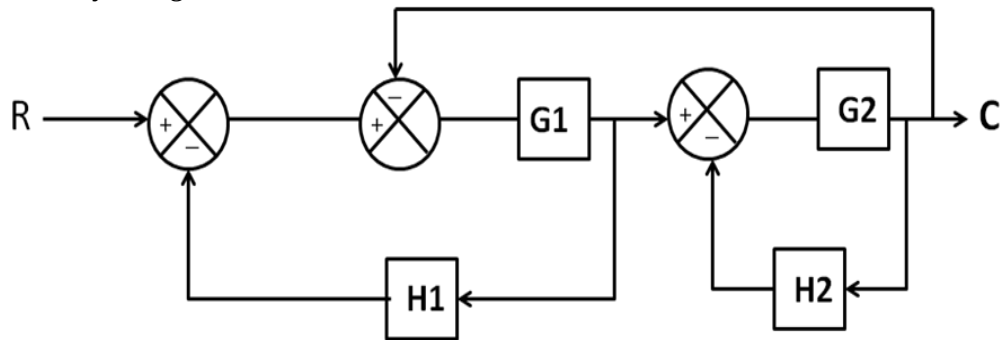
**(b) Answer any two questions.**

[10]

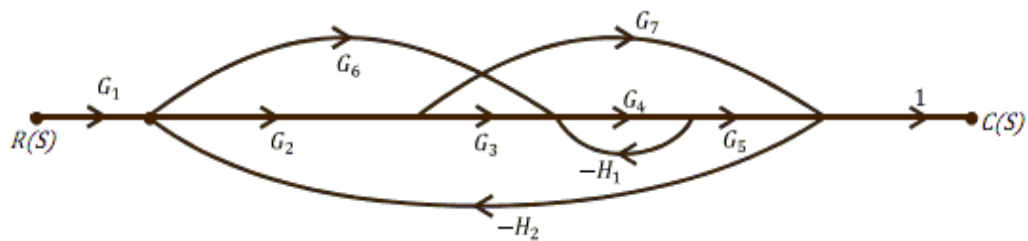
- 1) For the block diagram shown in figure below, determine the transfer function.



- 2) Using block diagram reduction technique, find the closed loop transfer function of a system given below.



- 3) Obtain transfer function from the signal flow graph shown in following figure (Use Mason's gain formula).



**Q - 3 (a) Answer any two questions.**

**[14]**

- 1) Consider a unity feedback control system with an open loop transfer function

$$G(s)H(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Draw the Root Locus for the system with the gain K as variable.

- 2) A Feedback control system has an open-loop transfer function

$$G(s)H(s) = \frac{K(s+2)}{(s+1)^2}$$

Sketch the Root Locus as K is varied from 0 to  $\infty$ .

- 3) A feedback Control System has an open loop transfer function

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Sketch the Root Locus as K is varied from 0 to  $\infty$ .

### SECTION – II

**Q – 4 (a)** The characteristics equation of a system is given by

**[03]**

$$3s^4 + 10s^3 + 5s^2 + 5s + 1 = 0$$

Determine the stability by means of Routh- Hurwitz criterion.

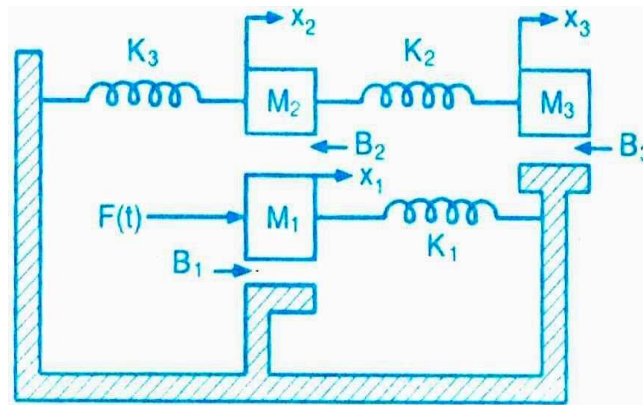
**(b)** Using Routh's criteria, for the unity feedback system,

**[04]**

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$$

Find range of values of K.

- Q - 5 (a)** For the following translation system shown in Figure, Obtain the differential equations and draw the electrical network using force-voltage analogy. [08]



- (b)** Sketch the asymptotic bode plot for the transfer function given below and also find [06]  
gain cross over frequency, phase cross over frequency, gain margin and phase margin  
of the system. Comment on Stability

$$G(s)H(s) = \frac{100}{s(s+1)(s+2)}$$

**OR**

- (b)** Define following with the help of neat diagram. [06]

1. Gain cross-over frequency
2. Phase cross-over frequency
3. Gain cross-over
4. Phase cross-over
5. Phase margin
6. Gain Margin

- Q - 6 (a)** Answer any one question. [08]

- 1) Using Nyquist criterion, examine the closed loop stability of a system whose open loop transfer function is

$$G(s)H(s) = \frac{2500}{s^3(s+300)}$$

- 2) Sketch the Nyquist plot for a system with

$$G(s)H(s) = \frac{1}{(s+3)(s+2)}$$

Comment on Stability.

**(b) Answer any two questions.**

**[06]**

- 1) A unity feedback system has open loop transfer function given below

$$G(s) = \frac{10(s+1)}{s^2(s+2)(s+10)}$$

Determine

- i. Characteristics equation.
  - ii. Natural frequency of oscillations
  - iii. Damping Ration
- 2) A unity feedback system has an open loop transfer function given below:

$$G(s) = \frac{(1+0.4s)}{s(s+0.6)}$$

- i. Obtain Unit step response.
  - ii. What is Rise time?
  - iii. What is peak overshoot?
- 3) A unity feedback system has

$$G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$$

Determine

- i. Type of system
- ii. All error coefficients
- iii. Error for ramp input with magnitude 4.

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